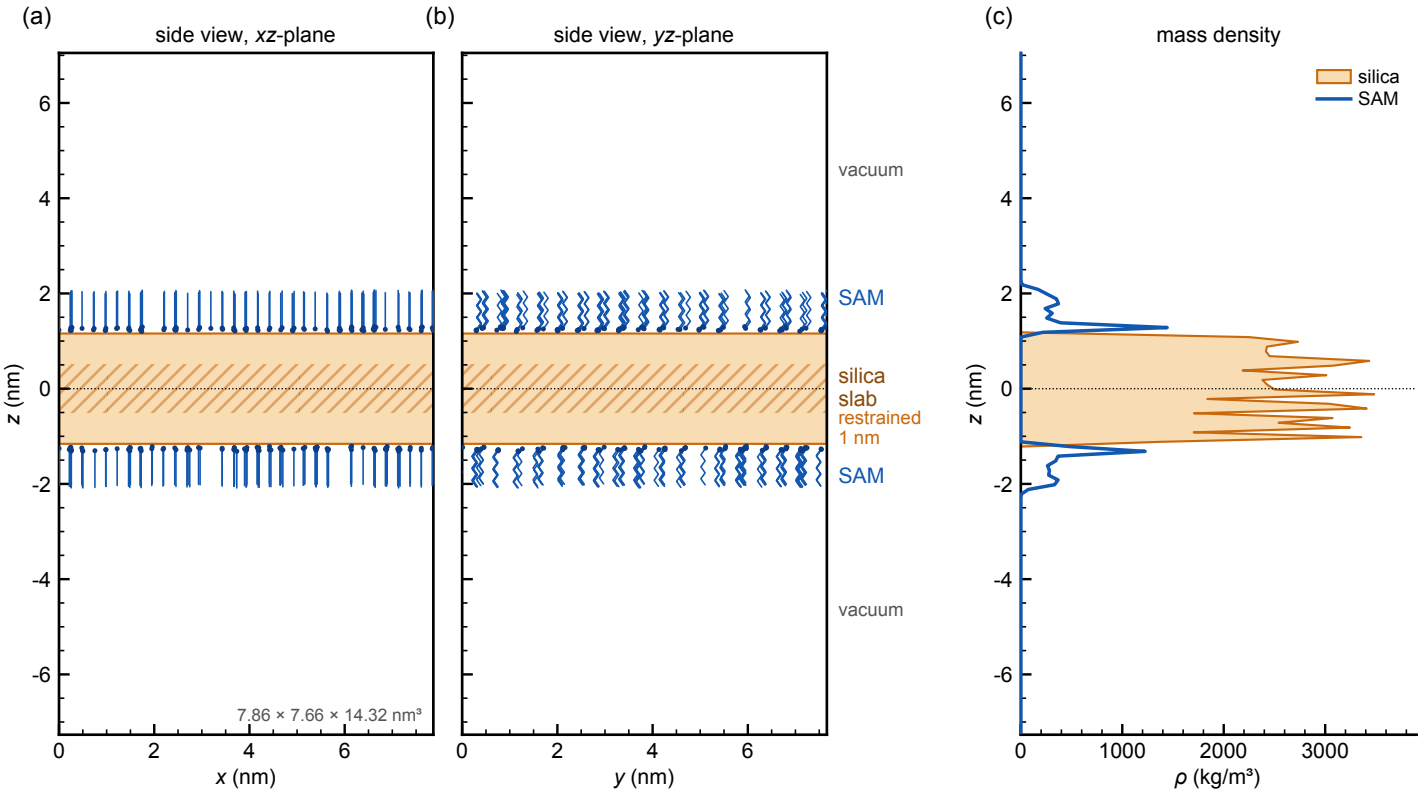
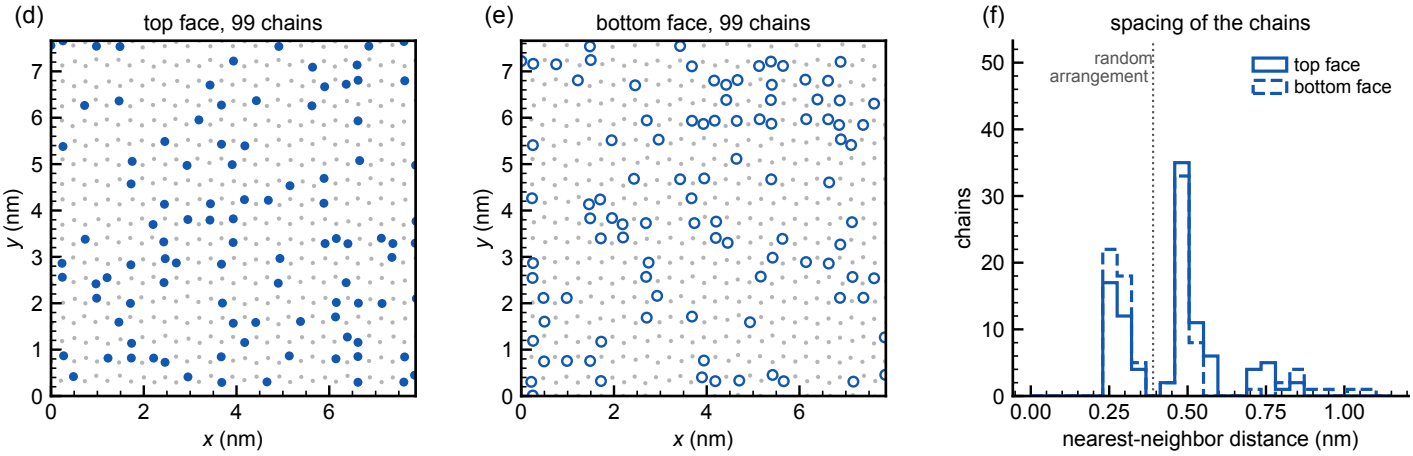
SAM system, L = 6, Γ = 1.65 molecules/nm²

Alkylsilane SAMs on both exposed (001) faces of a hydroxylated α-quartz slab. Both faces have the target surface coverage; the binding locations of the two faces were chosen independently.

build settings: surface_mode = slab x_tiles = 1 y_tiles = 1 random_seed = 1 system_name = func





● binding location, top face ○ binding location, bottom face • free hydroxyl oxygen

System

Quantity	Value
Chain length L	6 carbon atoms
Target coverage Γ	1.65 molecules/nm²
Chains, top / bottom face	99 / 99
Coverage, top / bottom face	1.644 / 1.644 molecules/nm²
Free OH groups, top / bottom face	7.92 / 7.92 per nm²
Simulation box	7.8623 × 7.6601 × 14.3170 nm³
Face area	60.226 nm²
Silica slab thickness	2.32 nm
SAM thickness, top / bottom face	0.92 / 0.93 nm (all-trans)
Atoms: silica / chains / total	11898 / 4752 / 16650
Restrained centermost layer	4911 atoms (1 nm)
Slab center in the GROMACS box	z = 7.2665 nm

Simulation workflow

run: bash submit_all.sh

State	Stage folder	Run	dt (fs)	ns	Restrained
dry	1_minimization	minimization	-	-	whole slab
dry	2_nvt_dt0.25fs	NVT, V-rescale	0.25	1	whole slab
dry	3_nvt_dt0.5fs	NVT, V-rescale	0.5	1	whole slab
dry	4_nvt_dt1fs	NVT, V-rescale	1	1	whole slab
dry	5_nvt_equilibration	NVT, V-rescale	2	20	middle 1 nm
dry	6_production_nvt	NVT, V-rescale	2	100	middle 1 nm
wet	1_solvation	gmx solvate, TIP3P	-	-	-
wet	2_minimization	minimization	-	-	middle 1 nm
wet	3_nvt_dt0.25fs	NVT, V-rescale	0.25	1	middle 1 nm
wet	4_nvt_dt0.5fs	NVT, V-rescale	0.5	1	middle 1 nm
wet	5_nvt_dt1fs	NVT, V-rescale	1	1	middle 1 nm
wet	6_npt_crescale	NPT, C-rescale	1	10	middle 1 nm
wet	7_nvt_equilibration	NVT, V-rescale	2	20	middle 1 nm
wet	8_production_npt	NPT, P-R	2	100	middle 1 nm

Lengths in nm; z = 0 at the center of the silica slab (z in the GROMACS files = z + 7.2665 nm).